

The economics of hybrid power systems for sustainable desert agriculture in Egypt

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Abstract

Egypt has embarked on an ambitious desert land reclamation program in order to increase total food production. Energy planners for these desert agriculture locations have chosen diesel generation power technology because minimization of the initial capital cost of a power supply system is their top priority. This heavy reliance on diesel generation has negative effects on the surrounding environment including soil, groundwater, and air pollution. Although good solar and wind resource prospects exist for the use of cleaner hybrid power systems in certain desert locations, little research has been done to investigate the economic potential of such systems in Egypt's desert agriculture sector. Using optimization software, we assess the economics of hybrid power systems versus the present diesel generation technology in a remote agricultural development area. We also consider the emission reduction advantages of using hybrid systems. Interestingly enough, optimization results show that hybrid systems are less costly than diesel generation from a net present cost perspective even with the high diesel fuel price subsidies. Since hybrids are also more environmentally friendly, they represent a strong step towards achieving sustainable desert agriculture.

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1. Introduction

Since the mid-1980s, Egypt has embarked on an ambitious scheme to increase domestic food production through desert reclamation. Since the inception of this program, about 2 million acres of desert land has been reclaimed by both public and private sector companies. The government has a target of 3.7 million acres to be achieved by 2017. Successful desert reclamation requires three key ingredients: good quality top soil, adequate water, and reliable electricity supply [1].

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The focus of this paper is on the third ingredient, which is the electricity supply. Presently, the power supply of choice in land reclamation in Egypt has usually been diesel generation. This has been the case for land areas located far off the grid, which comprises the majority of the land reclaimed. Only users on lands close to the grid have been connected to it. The fact that diesel fuel is heavily subsidized by the government has also contributed to the strong dependence on diesel generation in remote area agricultural projects. However, the widespread use of diesel generators in these agricultural areas is currently posing a threat to the environment. Soil pollution occurs when diesel spills take place and the disposal of burned diesel engine oil on the ground leads to groundwater pollution. Finally, air pollution increases as emissions of soot and green house gases (GHG) from the diesel generators continue unchecked. All these effects put the environmental sustainability of desert land reclamation in Egypt in jeopardy. The only way to solve this sustainability problem is to consider alternative sources of energy for these agricultural projects. It is the objective of this paper to analyze the economics of renewable energy technologies in remote area applications in Egypt given the current price and subsidy structures.

The potential for using clean energy technologies in remote desert agricultural areas is good given the abundant solar insolation in Egypt and/or wind resources. But up to now, the use of renewable energy has been confined to demonstration projects where single renewable energy technologies were used. No attempts for the adoption of large-scale renewables have been made so far. In addition, the use of hybrid power systems where more than one power generation technology is used has also not been applied.

2. Literature review

Very little has been written on applications and usages of hybrid power systems in Egypt. Renewable energy as a technology is still in its very early stages of adoption in Egypt, hence the scarcity of literature on this subject. However, adequate research has been done on the uses and applications of renewable energy technologies in the developing world in general. Baring-Gould [2] outlines the foundations of hybrid power systems architecture and design. He presents hybrid systems as an optimum approach for stand-alone power supply systems for certain remote area applications where the load demand and the available renewable energy resources are in-line with the basic requirements a hybrid system needs to be functional. He states that what really makes a hybrid system economical are the reductions in fuel costs, as well as reductions in operation and maintenance costs both achieved through the use of these systems. The key factor that leads to achieving these two cost reductions is the decreased operating hours of a diesel generator due to increased reliance on renewable energy resources [2]. Wichert et al. [3] presents an evaluation of the technical and economic characteristics of hybrid power systems, and outlines the expected future directions for the development of hybrids. They include a brief background description of situations where the use of diesel-only generation becomes uneconomic. He compares the total net present costs of hybrids versus diesel-only generation. Hybrids are in a more favorable position when the cost of diesel fuel transportation is incorporated in the analysis. Hence, hybrids are becoming increasingly popular among energy planners dealing with remote areas where a continuous and safe power supply is needed. This is happening in both developed and under-developed countries but at varying levels of adoption and deploy-

ment. They state that most developing countries are still suffering from lack of clear understanding of hybrids and their economics [3]. Hunter [4] presents a detailed account of the technical and operational characteristics of hybrid systems that comprise wind turbines and diesel generators. He states that among the various disadvantages of all-diesel generation in remote areas is that there has to be a stand-by diesel generator to be used only when repairs or maintenance are being done to the main diesel generators. This leads to excess capacity in the all-diesel generation system. The answer to elimination of this excess capacity is to introduce wind energy generation to the system. On the other hand, the author has presented wind-diesel hybrids as the ultimate solution without telling the reader how to deal with lack of adequate wind speed in some locations. Therefore, this study close this gap by looking at the potential for combining PV in a wind-diesel hybrid system is an issue that our study will tackle in the coming sections [4]. In a report by the Ministry of Agriculture (2000) in Cairo entitled *Desert Agricultural Investment Indicators*, a chapter is allocated for the potential uses of renewable energy technologies in desert agriculture in Egypt. This outlines the farm activities that could possibly be supplied with power by using renewable energy technologies. The report stresses the importance of accurate demand forecasting for each location before a renewable energy system is designed. Among the important variables to be taken into consideration are the land area, cropping patterns, and animal wastes that could be used in biomass power generation. The report mentions that the use of wind energy in the East Owienat region has good potential given the strong winds prevailing in the area. Power generated from wind turbines could be used in water pumping and load demand. Various sizes of wind farms could be built depending on the size of the land and the water needs for each land area [5]. El Hadidy and Shaahid, analyze the design and sizing of a hybrid system comprising wind turbines, a PV array, a battery bank for storage and a diesel generation back up system. The annual electrical energy demand of 41,531 kWh is supplied by the system. The system is designed and implemented in the western City of Dhahran in Saudi Arabia. The load demand derived in this paper is generated from the annual electricity consumption of a “198 square meter, four typical centrally air-conditioned family houses” located in Dhahran, Saudi Arabia. The system designed by the authors to satisfy the load demand comprises two 10 kW wind turbines and an array of PV with a total area of 120 m² [6]. Although none of the literature presented above deals specifically with the issue of using hybrid power systems for desert agriculture systems, some of the findings reached in these studies were instrumental in guiding this study in its course of analysis.

3. Case study

Our case study focuses on a newly developed desert agricultural area called East Oweinat located in the southwestern desert of Egypt. It is 380 km west of the nearest power grid line at Aswan City. The area gets its water from a renewable underground aquifer that is recharged from Lake Nasser. This area is the target for an agricultural development scheme that will result in the reclamation of about 200,000 acres. The development of about 10,000 acres is currently taking place where several diesel generators have been deployed by the government. These diesel generators are being used to supply power to submersible water pumps, irrigation systems, and farm workers' residences. A groundwater aquifer is the only source of water in this

area, and water pumping is the main source of electricity load demand. As mentioned above, the precarious and pristine environment of this location is being threatened by pollution caused by these diesel generators. Since the prospects for using clean energy technologies in this location are promising, particularly for hybrid power systems, we look at the economics of such systems. The size of the load demand in this area is not high enough to justify connecting this location with the national power grid, so we focus on the currently feasible components for a hybrid system: photovoltaic (PV), wind turbines, diesel generators, and battery banks.

The present diesel generation is designed for worst-case scenarios. Since such systems must be able to satisfy peak loads that may last for short period of times, they are characterized by huge excess capacity. In addition, having an all-diesel system implies that the diesel operates the whole year given the fact that there is a residential load that requires 100% system availability. This implies that most of the time the diesel generator is working at less than its efficient capacity levels. It also means that there are considerable GHG emissions and soot in the project location along with soil pollution, which seriously affects the sustainability of the surrounding natural resources.

In order to design a cost-minimizing power supply system, we need to consider the available wind and solar resources in the project area [7]. Fig. 1 presents the monthly solar resources in kWh/m²/day based on the latitudinal (25° N) position of the area. The solar data used are fairly precise and available from various sources. As for the wind resources data, finding accurate wind data has proven to be a challenge given the fact that the location is fairly new and there are no precise historical data for this location.

The wind data we use is based on estimates made by government energy planners, and international donor agencies that have surveyed the wind resources in the area. These data may be refined as more experience is gained in the planning area. For now, we use an average annual speed of 5.5 m/s, which is considered a modest estimate given the known strong sustainable wind in this location. Fig. 2 presents the daily wind speed profile in this location.

Accurately forecasting the load demand in the location is crucial for an optimal system design [8]. Two types of loads are identified and forecasted. The first type of load demand is that generated by housing units for farm workers. This primary load is forecasted based on load demand from similar locations in other developing countries. In this analysis, we use estimates based on 50 workers residing in the area and responsible for 100 acres. The load demand for the residential units used by these fifty workers is forecasted using a bottom up approach, which assumes basic electrical appliances in the dwellings.

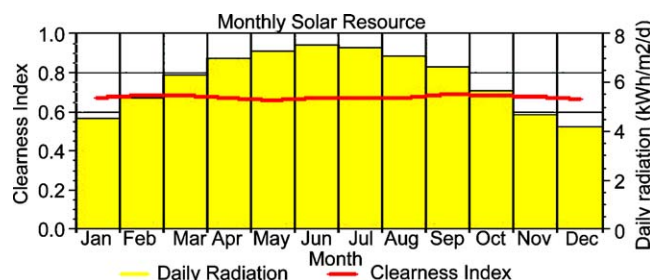


Fig. 1. Monthly solar resource.

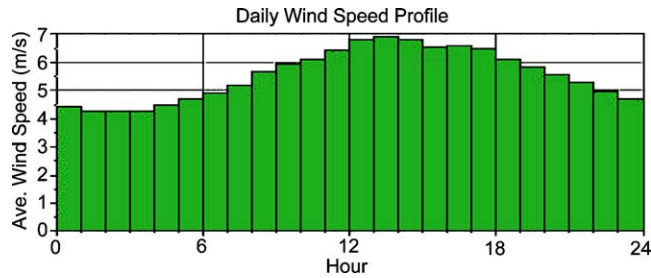


Fig. 2. Daily wind profile.

The second type of load demand is generated by water pumping and irrigation activities, which represents the bulk of total load demand in the targeted 100 acres farm. This load is a deferrable load because it can be satisfied during any time of the day and not in specific hours. Such a load characteristic offers the cost-minimizing advantage of satisfying the deferrable load whenever there is adequate wind and/or solar resources. The estimation of the deferrable load was based on the following assumptions:

- Two crops are being planted in the 100 acres, potatoes and corn.
- More than one crop is cultivated in order to benefit from differences in growth cycles that level the daily water requirements per acre, and hence level the load demand profile.
- The daily consumptive water requirement per acre of potatoes or corn is 30 cubic meters using sprinkler irrigation with 80% efficiency, and an evapotranspiration level for potatoes of 9 ml/day.
- The groundwater depth (static head) in the area is around 100 m.
- One groundwater well serves 100 acres¹ assuming drip or sprinkler irrigation is used.

It should be mentioned that specifying the type of crops is important for accurate load demand forecasting. Each crop has different water needs during its growth cycle, which directly affect the required amount of water pumping, and hence the load demands from the submersible pumps. Table 1 presents load figures for both types of loads.

The quantity of groundwater pumped to the surface is translated into power needs using the following equation. The flow rate is based on 12 h daily operation of the submersible pump.²

$$\text{Watts} = \frac{\text{Dynamic head} * \text{flow rate (liters/min)} * 16}{\text{Submersible pump efficiency}} \quad (1)$$

In this analysis, we use the current diesel fuel market price in Egypt which is a subsidized price of US\$0.09 per liter at a shadow exchange rate of US\$1 per 5.6 Egyptian pounds. This regulated diesel fuel price is paid by diesel fuel consumers in Egypt regardless of their location. In other words, the government is subsidizing the transportation cost incurred to transport this

¹ This is based on hydrological studies done by the Ministry of Agriculture, Cairo, estimating the maximum sustainable yield (MSY) from groundwater aquifers in this location.

² Submersible pump efficiency used in the calculation is 55%, and the figure of 16 in the equation is a constant.

Table 1
Project site load data

	Total annual load served (kWh)	Annual average (kWh/day)	Annual peak (kW)
Primary load	36,683	100.3	9.6
Deferrable load ^a	582,973	1591	133

^a The deferrable load figures are based on a submersible pump with a flow rate of 1.1 m³ per min (1100 l/min), 12 h of daily pump operation, static water head of 100 m, and pump efficiency of 55%.

fuel to East Owienat in addition to the overall subsidy received by all of Egypt. It should be mentioned that Egypt imports a large share of its domestically consumed diesel fuel due to capacity constraints in domestic refineries. Hence, assuming an approximate marginal cost price of diesel at approximately \$0.20/liter, which is how much the government pays to import diesel from international markets, and ignoring the cost of transporting the diesel fuel to the project's remote location, the subsidy paid by the government on diesel fuel is at least 50%. It should be mentioned that in year 2002, Egypt will spend US\$2.6 billion on subsidization of oil products sold in the domestic market, of which diesel fuel represents a large share. Any savings in the much-needed hard currency used to import this diesel fuel would have favorable budgetary and balance of payments effects at the national level.

4. Analysis results

HOMER (hybrid optimization model for electric renewables), a software developed by the US National Renewable Energy Laboratory, Colorado, is used to identify the cost-minimizing combination of power generating technologies that would suit the load profiles. The cost minimization model is as follows:

$$\begin{aligned} \text{Minimize annualized cost} &= \sum_{i=1}^n (K_i + RP_i + OM_i) \\ \text{Subject to } \sum (\text{load} - \text{power output}) &\leq \text{max unserved load (\%)} * \text{Load} \end{aligned} \quad (2)$$

where AC is the annualized cost that we want to minimize, K is the initial capital cost, RP is the replacement cost per power generating machine, and OM is the annual operation and maintenance cost per power generation machine. In this analysis, the annualized capital cost is calculated by evenly spreading the initial capital costs and the fixed operating costs of a component over a projects expected lifetime. In addition, the levelized cost of energy (LCOE), net present cost (NPC),³ and carbon emissions for the cost-minimizing hybrid system are also calculated.

In this optimization process, various sizes of power inverters, wind turbines, diesel generators, and batteries were included in the search space of the program. Several runs were made before determining the optimum sizes of equipment we needed to include to get the best results. Table 2 presents the different component sizes searched by the software. For example, there are five sizes of photovoltaic arrays, two sizes of wind turbines, which display some economies of scale if

³ This is calculated based on a 25 years project lifetime, 0% unserved load, and a real interest rate of 8%, a commonly used rate according to discussions with Egyptian government planners.

Table 2
Optimization search space for hybrid power systems

PV array (kW)	20 kW wind turbine (no. of units)	50 kW wind turbine (no. of units)	Diesel generator (kW)	Battery (kWh)	Inverter (kW)
1	1	1	50	30	5
5	2	2	100	40	10
10	3	3	120	50	15
15		4	200	60	20
20		5	250	70	30
		6		80	
		7			

more than one is purchased at a time, six sizes of battery, and five sizes of inverters to change DC to AC current.

HOMER then mixes and matches these components to find what combination satisfies the load requirements for the planning area at the lowest net present cost. A key objective is to compare the LCOE and the carbon emissions of the present situation diesel-only generation with those of various hybrid systems. The advantage HOMER offers is that it gives us the opportunity to design a cost-minimizing power system that provides a tailor-made power supply for the specific load demand. HOMER also minimizes the excess capacity problem mentioned above [9].

Three types of cost data for each component were used in the analysis. These are capital cost, replacement cost, and operation & maintenance cost. Capital cost is the estimated market price of the equipment in Egypt. The replacement cost is the capital cost of replacing a component at the end of its lifetime. This may be different from the initial capital cost because not all of the components may require replacement at the end of its life. For example, the wind turbine nacelle may need replacement but the tower may not. Operation and maintenance cost includes the cost of labor in Egypt. Table 3 presents these costs for the basic unit sizes. Based on these cost figures, the costs for units with larger rated capacities are extrapolated.

Optimizing hybrid power systems requires a careful choice of the dispatch strategy for the components of the system. Renewable energy is always used when available and any extra

Table 3
Cost figures used for the various components of hybrid systems

Rated capacity	Capital cost (\$)	Replacement cost (\$)	Operation and maintenance cost
1 kW PV	8000	6000	\$100/year
20 kW wind turbine	55,000	45,000	\$800/year
50 kW wind turbine	80,000	70,000	\$1200/year
8 kW diesel generator	7000	6000	\$0.25/h ^a
1 kWh battery	100	90	\$5/year
4.8 kW inverter	3500	3400	\$40/year

^a An initial survey by the authors found that in Egypt, O&M cost for diesel generators is typically higher than manufacturer's estimates. This is because preventive and regular maintenance are largely ignored, as diesel generators are usually maintained only when there is a major breakdown which requires replacing vital and costly spare parts.

renewable power is used to charge the batteries unless they are fully charged. If more power is needed HOMER decides whether to run off of batteries or start up the diesel generators. Since diesel is expensive to start up and run at less than full capacity, it is only run when excess load is high enough for long enough to justify its use, or there is no available battery power to meet excess load. HOMER then allows either of two types of diesel load strategy or a combination of the two. A cycle charging strategy always runs the diesel generator at full capacity once started and any extra power is used to charge the batteries. A load flow strategy runs the diesel generator only enough to satisfy the current excess load. We use a mixed strategy, which allows HOMER to use diesel to charge the batteries or not, depending on what is optimal in terms of production cost and extending battery life. This mixed strategy gives better results compared to either single strategy, as it combines the best of each strategy [9].

Based on the load profile (available from the authors upon demand), available renewable resources, and capital costs of various technology components given in [Tables 2 and 3](#), HOMER identifies the nine cheapest systems. We compare the two cheapest systems II and III, as shown in [Table 3](#) with Power System I, which is the present situation. Power System I, which is all diesel, is ranked at 6 by HOMER, while Power System II, the wind-diesel option, is slightly cheaper than Power System III, which has a very small component of solar [7].

5. Findings

From the [Tables 4 and 5](#), we can draw a number of important conclusions:

- Power System I with diesel-only generation offers the cheapest initial capital cost, but it has the costliest NPC and LCOE due to O&M and diesel fuel costs. On the other hand, both hybrid systems II and III have higher Initial Capital Costs, but much lower NPC. These results are reflected in the low LCOE enjoyed by both hybrid systems. Hence, although hybrids require high initial capital, they offer the lowest LCOE due to considerable reductions in diesel fuel costs and strong reliance on renewables.
- Diesel fuel and O&M costs for diesel generators are the biggest contributor to the high NPC Power System I (diesel-only system). Only if these high operating costs are discounted using a 25% real discount rate would diesel become the cheapest fuel option.
- Even with the substantial subsidy on diesel fuel, the NPC and LCOE of operating a diesel-only power generation system is still considerably higher than the cheapest hybrid systems that include renewable energy components. Hence, a low capital cost for diesel generators, with some encouragement from the artificially low diesel prices, have made diesel-only generation economically attractive, which is contrary to reality once recurring long-term costs are considered.

Table 4
Operational data for diesel generator in the diesel-only system

Operational life	1.8 years
Minimum load	67.8 kW
Maximum load	76.6
Specific fuel usage	0.36 l/kWh

Table 5
Power systems optimization results

System architecture	Wind speed m/s	Fuel price \$/l	Levelized cost energy (\$/kWh)	Totalnet present cost (\$)	Initial capital cost (\$)	Carbon emissions (ton/yr)	Diesel fuel (l/yr)	Diesel gen (h/yr)	Diesel%	Wind %	PV %
Power system I [diesel-only] One 100 kW diesel generator	–	0.09	0.16	1,043,687	87,500	162.2	225,258	8760	100	00	00
Power System II [Hybrid] Four 50 kW wind turbines + 100 kW diesel generator + 5kW inverter + 70kW battery	5.5	0.09	0.10	671,275	388,146	43.2	59,948	1825	25.5	74.5	00
Power system III [Hybrid w/PV] 1 kW PV + four 50 kW wind turbines + 100 kW diesel generator + 5 kW inverter + 80 kW battery	5.5	0.09	0.11	675,682	397,146	41.8	58,053	1771	24.8	74.9	0.3

- If the energy planners' key objective is to minimize carbon emissions in this remote location for environmental sustainability purposes, then Power System III should be selected. It has a LCOE that is much lower than LCOE of Power System I and it is only slight more costly than Power System II. The inclusion of a photovoltaic component in the Power System III's architecture has contributed to a reduction in diesel operating hours, and consequently diesel fuel consumption. In other words, the renewables fraction has increased and hence carbon emissions have decreased.
- Although Power System II wins out, Power System III with a solar component is a close second. As little as a 10% decrease in the capital cost, replacement cost, and O&M cost of PV technology *ceteris paribus*, leads to PV penetration in the cheapest hybrid power supply option. An advantage to having even limited PV penetration is that it would serve an educational role giving planners and users experience with the PV option.
- Deployment and adoption of hybrid power systems on a large scale will result not only in positive environmental effects, but also will lead to considerable savings in hard currency spent on diesel fuel importation, as well as savings on money spent on the diesel fuel subsidy. In both hybrid systems, diesel fuel consumption is cut to around 23% of the diesel-only case. This reduction is attributed to increased renewables fraction, and to a more efficient running of the diesel generator from an optimal dispatch strategy.
- The use of initial capital cost as the primary decision criteria in remote area power generation systems is misleading and inappropriate. Carbon emissions, fuel consumption, and diesel

hours of operation needs to be incorporated by energy planners in Egypt dealing with remote area power supplies.

- More sustainable agriculture could be archived through a change in the way the economics of power supply alternatives are assessed. Relying on LCOE, NPC and carbon emissions as the key decision parameters instead of using initial capital cost will result in the selection of cleaner remote energy technologies. This would ultimately be a step forward towards environmentally sustainable desert agriculture.
- For sustainability to be achieved in developing countries, a certain level of awareness needs to be achieved among energy planners. It is obvious that the selection of a power generation technology plays a crucial role in the level of environmental sustainability to be attained. Setting the priorities for environmental sustainability versus minimization of initial capital costs should be revisited. Routine processes used by energy planners in Egypt when designing power supply options need to be reassessed based on the new global warming realities.

6. Conclusion

In this paper, we have used an optimization model developed by the US National Renewable Energy Laboratory, to evaluate hybrid electricity systems for remote desert areas in Egypt. We compare hybrid systems with the current policy of using diesel generators and subsidized diesel fuel for all electricity needs. We find that selecting diesel generation as the default power option for remote areas in Egypt and ignoring other economically viable energy generation technologies results in unfavorable economic and environmental effects, which ultimately reduces the potential for sustainable agriculture. We find diesel-only generation a less desirable economical alternative even without removing the diesel fuel subsidy. If the diesel subsidy were removed, the case would be even stronger for abandoning the diesel-only option. Furthermore, diesel does not become the best option unless unrealistically high discount rates are used to compute the net present costs.

Although there are higher capital costs for the hybrid systems, their low operating costs work in their favor. A diesel-wind hybrid is currently the most economical option. However, lowering photovoltaic's capital cost, replacement cost, and O&M cost by only 10% would cause their penetration into hybrid systems in remote areas in Egypt. This preference for the hybrid systems is found even without including the favorable environmental benefits. Including these externalities would, of course, favor the use of the hybrids even more.

We believe that three major factors have played a role in making government energy planners select diesel-only generation for this remote location. The first factor is the low initial capital cost of diesel generators. The second is the lack of awareness among energy planners of alternative, environmentally friendly and cost-effective clean energy technologies. And finally there is the subsidy on diesel fuel, which although it decreases cost does not make diesel generation the most economical option. This paper provides evidence that the lowest initial capital cost is apparently the main factor that energy planners are concerned about and what they constantly attempt to minimize. If there are no incentives by the government to promote clean energy technologies, we do not expect to see rapid deployment of clean energy technologies in Egypt.

Therefore it is paramount for the government to develop policies to promote hybrid power systems and discourage diesel-only generation.

This research triggers many questions that are worth further research [10]. For example, it would be useful for the energy planners in Egypt to know what is the breakeven wind speed at which the wind component in hybrid systems in this location becomes unviable. It would also be beneficial to learn more about the specific impact of the current diesel fuel subsidy on the architecture of the hybrid systems used here and consequently the level of carbon emissions. A final point that is worth investigating is the economic viability of adding an elevated water tank to be used for irrigation versus direct pumping, and to what extent this tank affects the size of the battery banks used in the systems. We feel that providing answers to these and other prudent questions is a key ingredient for designing a desert development policy that aims at achieving sustainable desert agriculture in Egypt.

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